

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Semester Examination – Summer 2022

Course: B. Tech.

Branch : Electrical

Semester : IV

Subject Code & Name: BTEEPF405B/ Signals and System

Max Marks: 60

Date: 27.08.2022

Duration: 3.45 Hrs.

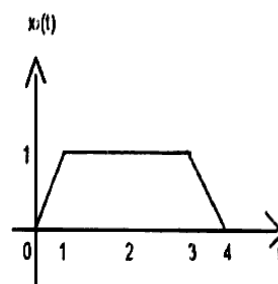
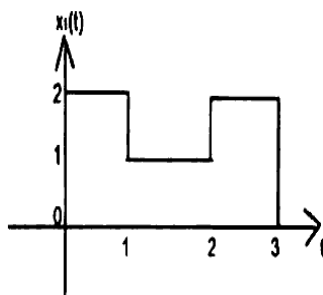
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) Plot the following signals: Create 6
- (I) $e^{-|t|}$ (II) $r(t) + \delta(t-2)$ (III) $\Lambda(t/1)$
- B) $x_1(t)$ and $x_2(t)$ are shown in following figures: Evaluate 6



Find:

- (I) $x_1(t) + x_2(t)$
- (ii) $x_1(t) - x_2(t)$
- C) If $X[n] = \{1, 2, 3, 4, 1\}$, then: Find $x[2n-3]$ and plot $x[2n/3]$ and plot $x[-2n]$. Evaluate 6

Q.2 Solve Any Two of the following.

- A) Define systems. Write the properties of the systems. Understand 6
- B) Check the following systems for Static/Dynamic/Causal/Non-Causal Analyze 6
- (i) $y[n] = 2x[n+1] + 3x[n] - x[n-1]$
- (ii) $y(t) = \frac{dx(t)}{dx} + x^2(t) + x(t+1)$
- C) By using the conventional approach check the causality and time invariant property of the following systems: Analyze 6
- (i) $y(t) = e^{x(t)} + x(t-1)$
- (ii) $y(t) = tx(2t) + x(t) + u(t-1)$

Q.3 Solve Any Two of the following.

- A) Find the convolution of following signals Evaluate 6

(i) $x(t) = \text{rect}(t/T)$ and $y(t) = u(t)$

(ii) $x(t) = u(t+5)$ and $y(t) = \delta(t-7)$

B) How the causality and stability of the system can be checked by impulse response of the system explain in detail. Understand 6

C) The response of discrete time LTI system is $\{2,4\}$ when the input is step. Analyze 6
Find the response of the system for the input $\{1,3,3\}$.

Q.4 Solve Any Two of the following.

A) Consider a Continuous time LTI system for which input $x(t)$ and output $y(t)$ are related by the following differential equation Evaluate 6

$$\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

Determine $h(t)$

(i) When system is Causal

(ii) When the system is stable

B) Find the Laplace transform of following signals and define ROC. Evaluate 6

(i) $x(t) = u(t) \cdot e^{(1-j)t}$

(ii) $x(t) = \sin 2t \cdot e^{2t} \cdot u(t)$

(iii) $x(t) = u(t) - u(t-1)$

C) Find Z- transform of following signals, and comment on ROC. Evaluate 6

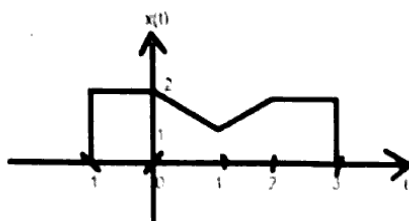
(i) $x[n] = \{1,2,3\}$

(ii) $x[n] = 3^n u[n] - 2^n u[n-1]$

(iii) $x[n] = (1/2)^n u[n]$

Q.5 Solve Any Two of the following.

A) For $x(t)$ shown in figure: Evaluate 6



Find:

(i) $\int_{-\infty}^{\infty} X(\omega) d\omega$

(ii) $\int_{-\infty}^{\infty} X(\omega) e^{j4\omega} d\omega$

B) Write the properties of Fourier transform. Remember 6

C) Derive the relation between trigonometric, exponential and polar Fourier series coefficient. Create 6

*** End ***